M00 Learner Book

Computer, Setup & Learning Foundations

Release Candidate 1 - Beginner-first production test module

Beginner Sufficiency Rule

This book is written so that a complete beginner can learn M00 without needing an outside tutorial. Videos, when added later, are support - not a replacement for the written explanation.

How to use this book

- Read the explanation before copying steps.
- Do the Try It sections on your PC.
- Do not worry about memorizing every command. Understand what it changes.
- When something fails, read the error and compare it with the troubleshooting section before starting over.

M00-L01 - Files, Folders, Drives and Paths

Learning objective

Understand where files live, how folders organize them, and how to describe a file location precisely.

Estimated focused time: 60-90 minutes including practice

1. The mental model

A file is one saved item, such as a spreadsheet, picture, Python program or PDF. A folder is a container used to organize files and other folders. A drive is a larger storage location. On most Windows computers the main drive is called C:.

Thing	Example	Meaning
Drive	C:	A large storage location
Folder	C:\Users\Student\Documents	A container
File	orders.csv	One saved item
Path	C:\Users\Student\Documents\orders.csv	The location of the file

A path is like an address. If someone tells you only "orders.csv", you know the file name but not where it lives. If they tell you the full path, you know the exact location.

2. Nested folders

Folders can contain folders. This is normal in technical projects. For example:

```
DE_COURSE/
  PROJECTS/
    sales_pipeline/
      data/
        orders.csv
      src/
        pipeline.py
```

Here, orders.csv and pipeline.py are in different subfolders even though they belong to the same project.

3. Absolute and relative location - first exposure

An absolute path starts from a fixed top-level location, such as C:\Users\Student\DE_COURSE\data\orders.csv. A relative path describes a location from where you are currently working, such as data\orders.csv. You only need the idea now; Linux and Python will revisit it later.

Try it now

1. Open File Explorer.

2. Open Documents.
3. Create a folder named DE_COURSE_PRACTICE.
4. Inside it create a folder named files_lab.
5. Create a text file named location_test.txt.
6. Click the File Explorer address bar and observe the location.

Common beginner mistakes

- Saving a file in Downloads and later assuming it is in Documents.
- Creating several folders with almost the same name, such as project, project2, project-final and project-final-new.
- Changing a file name but not noticing which folder contains it.

Check yourself

[ ] Can I explain the difference between a file and a folder?

[ ] Can I find the exact folder containing a file?

[ ] Can I describe what a path is in my own words?

M00-L02 - File Extensions and Common Data Files

Learning objective

Recognize the common file types used in this course and choose a sensible program for opening them.

Estimated focused time: 45-60 minutes

The extension is the ending after the final dot in a file name. It usually tells you the file format. Windows may hide known extensions by default, which can make beginner mistakes harder to notice.

Extension	Typical use in this course	Usually opened with
.xlsx	Excel workbook	Microsoft Excel or compatible spreadsheet app
.csv	Plain-text table	Excel, text editor, Python, database tools
.json	Structured text data	VS Code / text editor / Python
.txt	Plain text	Text editor
.sql	SQL script	VS Code or database client
.py	Python source code	VS Code / Python
.zip	Compressed archive	File Explorer / archive tool
.pdf	Fixed-layout document	PDF reader/browser

CSV is not an Excel workbook

A CSV can open in Excel, but it does not contain normal workbook features such as multiple sheets, formulas, formatting rules or PivotTables. It is fundamentally a text-based table.

Showing extensions in Windows

In current Windows File Explorer, use the View/Show options to make file name extensions visible. Exact menus can vary slightly by Windows build. The goal is simply to see the real file ending.

Try it now

1. Open the supplied practice_files folder.
2. Identify each file by extension before opening it.
3. Open sample.csv in a text editor first, then in Excel if available. Compare what you see.
4. Open sample.json in VS Code or another text editor and notice that it is readable text.

Common beginner mistakes

- Renaming report.csv to report.xlsx and assuming the file has been converted. Renaming changes the name, not the internal format.

- Opening a .py file in a word processor and adding formatting that breaks the code.
- Hiding extensions and accidentally creating names such as report.csv.csv.

Check yourself

[ ] Can I identify XLSX, CSV, JSON, SQL, PY and ZIP files?

[ ] Do I understand why changing an extension is not the same as converting a file?

M00-L03 - Download, ZIP, Extract, Copy, Move and Rename

Learning objective

Handle downloaded course materials without losing or overwriting files.

Estimated focused time: 60-90 minutes

A ZIP file is a compressed container. Many files and folders can be packed into one ZIP for downloading. You normally extract it before studying or running code.

Copy vs move

Action	Original remains?	Use when
Copy	Yes	You want a second copy, such as backup
Move	No, location changes	You are organizing the original into the right folder

A safe download workflow

1. Download the file.
2. Find it in Downloads.
3. Check the name and extension.
4. If it is a ZIP, extract it to a normal folder.
5. Move the extracted study folder into your course workspace.
6. Keep or remove the original ZIP according to the lab instruction - do not study directly inside the archive.

Why studying inside a ZIP is risky

Some applications let you preview files inside a ZIP. It can look as if you opened a normal folder, but saving changes can become confusing. Extract first unless the instruction explicitly says otherwise.

Try it now

1. Find DOWNLOAD_ME_FIRST.zip in this module's practice files.
2. Extract it.
3. Confirm that the extracted folder contains data, code and notes subfolders.
4. Rename the extracted folder to m00_download_lab.
5. Move it into DE_COURSE_PRACTICE.

Common beginner mistakes

- Opening files directly inside the ZIP and later wondering where changes were saved.
- Using Cut/Move when you intended to keep an untouched original.
- Adding words such as FINAL and FINAL2 repeatedly instead of using a stable naming system.

Check yourself

[ ] Can I explain what extracting a ZIP does?

[ ] Can I distinguish copy from move?

[ ] Can I find a downloaded file without downloading it again?

M00-L04 - Installing Software Safely

Learning objective

Understand installers, official download sources, versions and basic verification.

Estimated focused time: 45-60 minutes plus any installation time

Installing a program usually means running an installer that copies program files, creates settings and may add shortcuts or command-line access. A downloaded installer is not the same thing as the installed application.

Safe installation checklist

- Prefer the official product website or official app store.
- Check whether the instructions require Windows 64-bit or another platform.
- Do not install random "download manager" programs from advertisement pages.
- Read the installer screen instead of clicking Next automatically.
- After installation, launch the program once and verify the version when the course asks for it.

What is PATH?

PATH is a list of folders your terminal searches when you type a command such as python. You do not need to edit PATH manually in M00. You only need to understand why a program can be installed yet a terminal command might still say it cannot find it.

Verification practice

1. Open VS Code if it is already installed.

2. Find Help > About (or the equivalent version screen).

3. Write the installed version in your completion record.

4. If VS Code is not installed, use the official installation route specified by the course setup instructions before continuing.

Common beginner mistakes

- Downloading from an advertisement that imitates the real download button.
- Assuming the installer file itself is the installed application.
- Installing multiple versions without knowing which command is used.

Check yourself

[ ] Do I understand installer vs installed application?

[ ] Can I verify that an application launches successfully?

[ ] Do I know what PATH means at a basic level?

M00-L05 - VS Code From Zero

Learning objective

Open a project folder correctly, create/edit files, and understand the main VS Code areas used in the course.

Estimated focused time: 60-90 minutes

VS Code is a code editor. It is not Python, SQL or a database by itself. It gives you a convenient place to view project folders, edit files and open an integrated terminal.

Area	Purpose
Explorer	Shows files/folders in the opened workspace
Editor	Shows the file you are editing
Tabs	Switch between open files
Terminal	Runs commands from the current project environment
Extensions	Adds support for languages/tools when needed

Open the folder, not random files

For technical projects, use File > Open Folder and choose the project root. This gives VS Code the correct workspace context. Opening one random .py file is not the same thing.

Guided first workspace

1. Open VS Code.
2. Choose File > Open Folder.
3. Open DE_COURSE_PRACTICE\m00_download_lab.
4. In the Explorer panel open notes\instructions.txt.
5. Create a new file named evidence.txt in the project root.
6. Type: VS Code workspace opened successfully.
7. Save the file, close VS Code, reopen the same folder and confirm evidence.txt still exists.

Common beginner mistakes

- Opening a file but not the project folder.
- Typing into an unsaved tab called Untitled and then closing it.
- Saving a new file in an unexpected directory because the workspace was not opened correctly.

Check yourself

[ ] Can I open a folder as a VS Code workspace?

[ ] Can I create and save a new file in a specific folder?

[ ] Can I find the integrated terminal?

M00-L06 - Terminal Fundamentals

Learning objective

Understand the prompt, current directory, command and argument, then navigate folders using basic commands.

Estimated focused time: 75-90 minutes

A terminal lets you give text commands to a shell. The shell interprets those commands. Later tools such as Git, Python, Docker and many cloud utilities will use terminal commands, so M00 removes the fear early.

Four ideas before commands

- Prompt: the line where you type a command.
- Current directory: the folder the shell is currently working inside.
- Command: the action, such as cd.
- Argument: extra information given to a command, such as the folder name to enter.

Goal	Windows PowerShell / CMD style	Linux/WSL style
Show current location	pwd in PowerShell; cd in CMD	pwd
List contents	dir or ls in PowerShell	ls
Change folder	cd folder_name	cd folder_name
Create folder	mkdir folder_name	mkdir folder_name

Do not memorize both operating systems now. M06 will teach Linux/WSL properly. In M00, the key outcome is understanding current directory and navigation.

Terminal navigation lab

1. Open the VS Code terminal while m00_download_lab is open.
2. Display the current folder.
3. List files and folders.
4. Change into the data folder.
5. List its contents.
6. Return to the parent folder using cd .. .
7. Create a folder named terminal_created.
8. Confirm it appears in the VS Code Explorer.

Common beginner mistakes

- Typing a file path as if it were a command.
- Running a command from the wrong current directory.
- Using spaces in a path without appropriate quoting when the shell requires it.

- Deleting things with commands before understanding the current directory. M00 intentionally avoids destructive terminal commands.

Check yourself

[ ] Can I explain what current directory means?

[ ] Can I list a folder and change into a subfolder?

[ ] Can I return to the parent folder?

M00-L07 - Errors and Beginner Troubleshooting

Learning objective

Use a controlled troubleshooting process instead of random changes.

Estimated focused time: 60-90 minutes

An error message is information. It usually tells you what operation failed, where it failed, and sometimes why. Technical work becomes much easier when you stop treating every red message as a signal to restart everything.

The M00 troubleshooting loop

1. Reproduce the problem once.
2. Read the complete message.
3. Identify the file, path, command or program mentioned.
4. Form one simple hypothesis.
5. Change one thing.
6. Retry.
7. If it works, write what caused the problem. If not, undo or reassess rather than changing five more things.

Example: file not found

Suppose an instruction says to open data\orders.csv, but the file is actually in data\archive\orders.csv. Reinstalling VS Code will not help. The evidence points to a location problem.

Symptom	First thing to inspect
File not found	Current folder and exact path
Command not recognized	Is the program installed? Is PATH/config correct?
Permission denied	Are you allowed to access/modify this location?
Cannot open file	Correct application? Corrupt/incomplete file?
Unexpected file name	Extension visible? Duplicate extension?

Broken setup exercise

1. Open the supplied broken_setup folder.
2. Read TASK.txt.
3. Attempt the task exactly once.
4. Use the error/symptom to identify the wrong file location.
5. Repair only the location problem.
6. Write one sentence in repair_note.txt describing the cause.

Common beginner mistakes

- Changing several settings at once, so you never learn which change mattered.
- Searching the internet before reading the full error.
- Reinstalling software to fix what is actually a path or filename issue.

Check yourself

[ ] Can I describe the six-step troubleshooting loop?

[ ] When I see file-not-found, do I inspect the path before reinstalling anything?

M00-L08 - Study Evidence, Attempts and Review Discipline

Learning objective

Separate learning, independent attempts, review and retests so progress means real skill rather than copied completion.

Estimated focused time: 45-60 minutes

During the course you will often be allowed to use a solution after making an attempt. That only works if the original attempt is preserved. Otherwise it becomes impossible to tell what you could do independently.

Folder/state	Purpose
LEARNING	Notes, teacher examples, follow-with-me work
ATTEMPTS	Your independent work before review
REVIEW	Explanations/solutions opened after attempt
RETESTS	Fresh second proof after repair
EVIDENCE	Screenshots, outputs, logs or submitted artifacts

Wrong attempts are valuable

A saved wrong attempt tells the Mentor what needs repair. A copied correct answer with no original attempt hides the weakness and makes the next module harder.

The normal assessment flow

Attempt -> Save -> Submit/check -> Review -> Repair if needed -> Fresh retest

Create your evidence structure

1. Inside DE_COURSE_PRACTICE create attempts, review, retests and evidence folders.
2. Create a file attempts\m00_test_attempt.txt and type one short sentence.
3. Copy it to evidence only after saving the attempt.
4. Do not edit the original attempt after review; create a new retest file instead.

Common beginner mistakes

- Overwriting the first attempt after seeing the answer.
- Marking a skipped lesson as mastered.
- Saving files with names that do not show whether they are attempt, review or retest.

Check yourself

[ ] Can I explain why attempts are saved before review?

[ ] Do I know where a fresh retest belongs?

[ ] Do I understand that skipped is not the same as mastered?

M00 Summary

- Files live inside folders, and paths describe where they are.
- Extensions identify file formats; changing a name does not convert a file.
- Extract ZIP archives before normal study work.
- Open project folders, not random individual files, in VS Code.
- The terminal always works from a current directory.
- Errors are evidence; change one thing at a time.
- Independent attempts must be preserved before review.

Before independent practice

- You have created DE_COURSE_PRACTICE.
- You can find DOWNLOAD_ME_FIRST.zip and extract it.
- You can open the extracted project folder in VS Code.
- You can navigate into a subfolder and back using terminal commands.
- You created attempts/review/retests/evidence folders.